

Rohan Mehra

Graduate Student in Computer Science

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Education

- 2025 – 2026 **ISIMA, Université Clermont Auvergne**
MS in Computer Science
- 2021 – 2025 **Indian Institute of Science Education and Research (IISER) Bhopal**
BS in Electrical Engineering and Computer Science

Research Experience

- Mar. – Aug. 2026 **LIFAT, CNRS, Université de Tours, Intern**
○ Advised by Prof. Gilles Venturini as part of RFAI team.
📄 ○ Working on AI-assisted fine-grained species recognition of insects.
○ Proposed a specimen-level training and aggregation framework; manuscript under review [1].
- Sem-1 2025–26 & May – July 2024, 2025 **Institut Pascal, CNRS, Université Clermont Auvergne, Project Student & Intern (×2)**
○ Advised by Dr. Mathieu Labussière as part of ComSee team.
📄 ○ Worked on deep learning with SWIR sensor for road perception in harsh weather (see [2], [3], [4]).
○ Tasks include Sem. Segmentation (CNN- & ViT-based) and Gen. AI (GAN- & diffusion-based).
- Jan. – April 2025 **IISER Bhopal, BS Final Project**
○ Advised by Dr. Arpit Sharma at the Dept. of EECS.
📄 ○ Worked on using Zero-Shot Learning model for Sector Classification in CrowdRE data.
○ Used models including Sentence-BERT, All-MiniLM-L12, Bert4RE and BERTOverflow
- May – July 2023 **Indian Institute of Space Science and Technology, Intern**
○ Advised by Prof. Deepak Mishra as part of the CVVR Lab
📄 ○ Worked on robotics and implementation of SLAM through Kalman filter Methods.
○ Used Robot Operating System (ROS), RViz, and Gazebo.

Publications & Preprints

Also available on [Google Scholar](#)

- [1] 🔗 **Rohan Mehra**, Daphné Démazier, Barthélemy Serres, Sébastien Moreau, Gilles Venturini, “INSC-SPEC: Specimen-Level Insect Species Recognition from Multi-View Images and Videos”, **Under review at a Conference, 2026.**
- [2] 🔗 **Rohan Mehra**, Alexandre Riffard, Yannis Loumouamou, Mathieu Labussière, “Analysis of SWIR Imaging Detection Performance Under Adverse Environmental Conditions for Autonomous Driving Systems”, **Under review at IEEE Transactions on Intelligent Transportation Systems (T-ITS), 2026.**
- [3] 🔗 Yannis Loumouamou, Alexandre Riffard, **Rohan Mehra**, Mathieu Labussière, “Experimental study of YOLOv8 fine-tuning for RGB-SWIR multimodal road detection”, *Reconnaissance des Formes, Image, Apprentissage et Perception (RFIAP)*, 2026.
- [4] 🔗 **Rohan Mehra**, Alexandre Riffard, Mathieu Labussière, Pierre Duthon, Romuald Aufrère, “Would SWIR modality help for detection and segmentation in harsh weather conditions? An experimental study.”, *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025. [Oral]

Teaching Experience

- 2024 & 2023 **Basic Electronics, IISER Bhopal** ($\times 2$)
- Teaching Assistant for an undergraduate class of 180 students for 2024-25/23-24 Semester I.
 - Conducted tutorial sessions (1 hour/week) and prepared assignment solutions.

Academic Service

Reviewer [2026] IEEE Sensors Journal, ECCV WiCV Workshop

Technical Skills

Programming Python, C++, C, Bash (Shell scripting)
AI/ML PyTorch, TensorBoard, Hugging Face, DL/CV Frameworks
Robotics ROS, Gazebo, RViz
Tools Git, Docker, $\text{\LaTeX}$, MATLAB, Wolfram Mathematica, Remote GPU Servers

Events

- Mar. 2026 **ELLIS Winter School on Foundation Models (FoMo)**
Selected to attend the FoMo Winter School hosted by the ELLIS unit Amsterdam, among a global pool of applicants (~ 250 participants).
- Feb. 2026 **Researchers' Day – DATA Program I-SITE CAP 20-25**
Presented a poster at PhD day, Clermont-Ferrand (sole master student).

Fellowships & Achievements

- 2025 **MIAI Cluster Excellence Scholarship**
Selected to receive a scholarship offered by the Multidisciplinary Institute in Artificial Intelligence (MIAI) in France to pursue master's program, awarded to ~ 20 candidates university wide.
- 2024 **IASc-INSA-NASI Summer Research Fellowship** [Declined]
Selected for research internship fellowship jointly offered by the Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, awarded to the top $\sim 5\%$ of applicants.
- 2021 **Indian Maritime University Undergraduate Admissions**
Secured All India Rank 9 out of 14,108 ranked candidates, placing in the top 0.1% nationwide.

Positions of Responsibility - Extra Curriculars

- 2023 – 2024 **Cultural Secretary, IISER Bhopal**
Elected representative of the student body for organizing all cultural events at IISER Bhopal. Led a team of 70 performers to the the Inter IISER Cultural Meet 2023 held at IISER Mohali.
- 2022 **NCC 'B' Certificate, National Cadet Corps, India**
Completed 'B' division NCC military training from 8 UP BN NCC, Aligarh.

Other

Languages English (fluent, TOEFL 99/120), Hindi (Native), French (Basic)
Hobbies Table Tennis, Quizzing